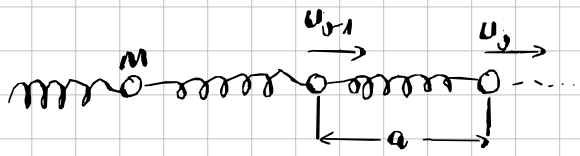
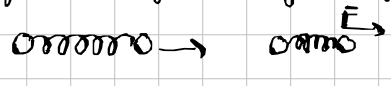
Phonon

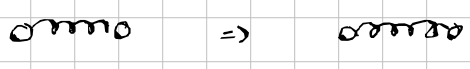


Consider a one dimensional chain of identical atoms each with mass M. separated by a
Equilibrium position $x_j^{(0)} = ja$ The movement is then $x_j(t) = ja + u_j(t)$

Each atom has a spring connecting, if a right neighbor stays still, the spring gets compressed



Then there is a force pushing back. If both atoms move right exactly



The spring is not stretch or compressed. The force is determined by

$$u_{j+1} - u_j$$

Similarly, we also have the left spring

$$M\ddot{u} = f(u_{j+1} - u_j) - f(u_j - u_{j-1})$$
$$= f(u_{j+1} + u_{j-1} - 2u_j)$$

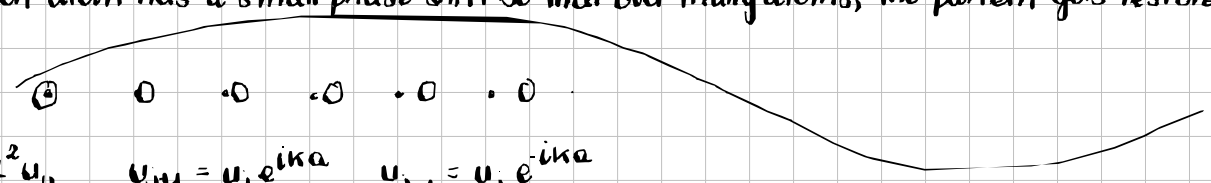
An atom does not feel a restoring force merely because it moves. It feels a restoring force because it moves differently from its neighbors

Using Ansatz, $u_j(t) = Ae^{-i(\Omega t - \kappa x_j)}$ $x_j = ja$

κ shows how each atom move relative to each other. For example, $\kappa = 0$ means all atoms move together when $\kappa = \pi/a$, adjacent atoms move in opposite direction

$\kappa = \pi/a$ O→ ←O $\kappa = 0$ O→ O→

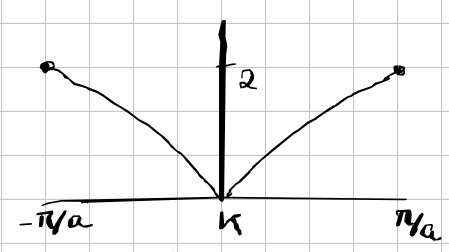
When κ is small, each atom has a small phase shift so that over many atoms, the pattern gets restored



$$\ddot{u}_j = -\Omega^2 u_j \qquad u_{j+1} = u_j e^{ika} \qquad u_{j-1} = u_j e^{-ika}$$

The equation gives

$$-M\Omega^2 u_j = f(u_j e^{ika} + u_j e^{-ika} - 2u_j)$$
$$\Rightarrow -\Omega^2 = \frac{f}{M}(e^{ika} + e^{-ika} - 2)$$
$$\Omega^2 = \frac{2f}{M}[1 - \cos(\kappa a)]$$
$$\Omega(\kappa) = 2\sqrt{\frac{f}{M}}\left|\sin\left(\frac{\kappa a}{2}\right)\right| \qquad 1 - \cos(\kappa a) = 2\sin^2\left(\frac{\kappa a}{2}\right)$$



This is phonon dispersion relation. Since $\kappa + 2\pi/a$ produces the same result. Only the first Brillouin zone is needed

$$u_j' = Ae^{i[(\kappa + 2\pi/a)ja - \Omega t]} = u_j e^{i2\pi j}$$

Case A: $\kappa = 0$ - all atoms move together so technically this is a translation rather than an oscillation

$$\Omega(0) = 0$$

CaseB: $\kappa = \pi/a$ - neighboring atoms move opposite

$$u_j(t) = A(-1)^j \cos(\Omega t) \qquad \Omega(\pi/a) = 2\sqrt{f/M}$$

The atoms osallate strongest.

Near $\kappa = 0$, $\sin\frac{\kappa a}{2} \approx \frac{\kappa a}{2}$, then $\Omega(\kappa) = a\sqrt{\frac{f}{M}}|\kappa|$

The group velocity

$$v_g = \frac{d\Omega}{d\kappa} \approx a\sqrt{\frac{f}{M}}$$ is the sound speed

f determines how quickly a disturbance is transmitted and M determines how strongly atoms resist acceleration

For $0 < \kappa < \pi/a$, $v_g(\kappa) = a\sqrt{\frac{f}{M}}\cos\left(\frac{\kappa a}{2}\right)$



The atoms are not literally connected by mechanical springs, they are interatomic potential energy

For Two atoms, the potential energy depends on separation r $E(r)$

At equilibrium, a_0, energy is minimized

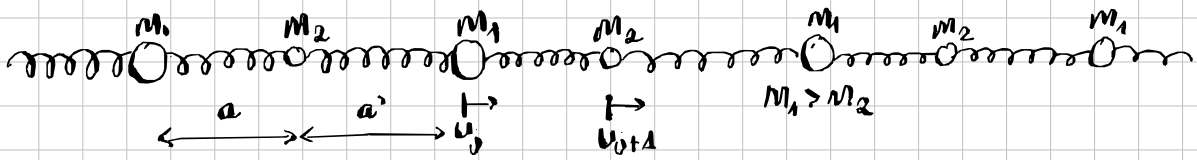
$$E'(a_0) = 0 \qquad E''(a_0) > 0$$

Near equilibrium, Taylor expansion gives

$$E(r) \approx E(a_0) + \frac{1}{2}E''(a_0)(r - a_0)^2 \qquad f = E''(a_0)$$
$$\approx E(a_0) + \frac{1}{2}f(r - a_0)^2$$
$$F = -\frac{dE}{dr} \approx -f(r - a_0)$$

This is Hooke's law

Diatomic 1D Lattice



For simplicity, Let $a = a'$ and $f = f'$ so that a unit cell is $2a$ $\Rightarrow G = \frac{2\pi}{2a} = \pi/a$ $\Rightarrow -\frac{\pi}{2a} \leq \kappa \leq \frac{\pi}{2a}$

Using Ansatz: $u_{1,n}(t) = A_1 \exp(i(2n\kappa a - \Omega t))$ n labels unit cell

$u_{2n}(t) = A_2 \exp(i(2n+1)\kappa a - \Omega t)$

Since a heavy atom and a light atom experience different accelerations under the same force. The amplitude must be determined separately.

The two Newton's equations become: